Problem No: 03

Problem Name: Implementation of Image Enhancement Technique: Brightness Enhancement

Objectives:

1. To understand the concept of image brightness enhancement.
2. To increase the brightness of an image using pixel intensity manipulation.
3. To observe the effect of brightness enhancement on image quality.

Theory: Brightness enhancement is an image processing technique used to increase or decrease the intensity of pixels in an image. It improves image visibility, especially for dark images.

The brightness enhancement operation is defined as:

$$s = r + k$$

Where:

- r= Input pixel value
- s= Output pixel value
- k= Brightness constant

If $k > 0$, the image becomes brighter.
If $k < 0$, the image becomes darker.

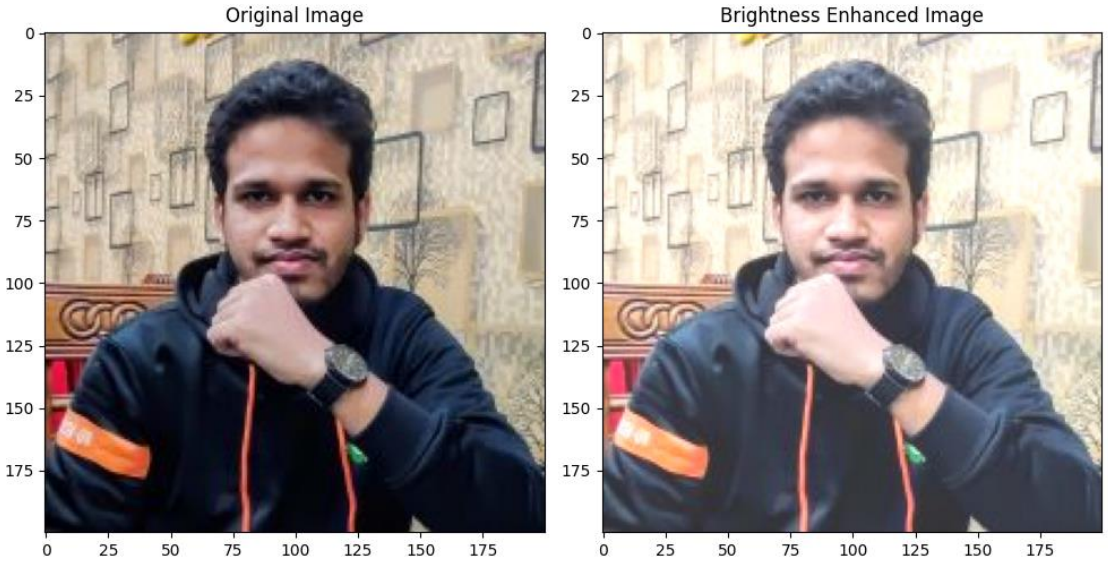
Program:

```
import cv2
import numpy as np
import matplotlib.pyplot as plt

img = cv2.imread('sm_pic.jpg')
brightness = 50
bright_img = cv2.convertScaleAbs(img, beta=brightness)
titles = ['Original Image', 'Brightness Enhanced Image']
images = [img, bright_img]
plt.figure(figsize=(10, 5))
for i in range(2):
    plt.subplot(1, 2, i + 1)
    plt.imshow(cv2.cvtColor(images[i], cv2.COLOR_BGR2RGB))
    plt.title(titles[i])

plt.tight_layout()
plt.show()
```

Output:

```
Brightness Enhanced Image Matrix:

[[[  8   1 239]
  [  9   2 240]
  [  0 247 232]
  ...
  [228 210 196]
  [228 210 196]
  [245 227 213]]

 [[ 16   9 247]
  [224 217 199]
  [202 193 178]
  ...
  [ 10 249 233]
  [ 15 253 239]
  [  2 242 226]]

 [[250 241 224]
  [219 210 193]
  [222 213 198]
  ...
  [  1 240 222]
  [  5 245 229]
  [  1 244 225]]

 ...
```

```
...

[[ 53  52  58]
 [ 52  51  57]
 [ 51  51  59]
 ...
 [ 56  62  76]
 [ 57  63  77]
 [ 58  64  78]]

[[ 65  64  72]
 [ 65  64  72]
 [ 62  62  70]
 ...
 [ 54  60  72]
 [ 55  61  73]
 [ 55  61  73]]

[[ 86  85  93]
 [ 85  84  92]
 [ 80  80  88]
 ...
 [ 54  60  72]
 [ 53  59  71]
 [ 54  60  72]]]
```



Result & Discussion:

The brightness enhancement technique successfully increased the visibility of the dark regions in the image. After adding a brightness value, the image appeared clearer and more visually enhanced.

Analysis:

Brightness enhancement works by increasing pixel intensity values. A higher brightness constant produces a brighter image, but excessive brightness may reduce image details due to overexposure.

Opinion:

Through this experiment, I learned how brightness enhancement improves image visibility by manipulating pixel intensity values. This technique is useful in photography, medical imaging, and digital image processing applications.